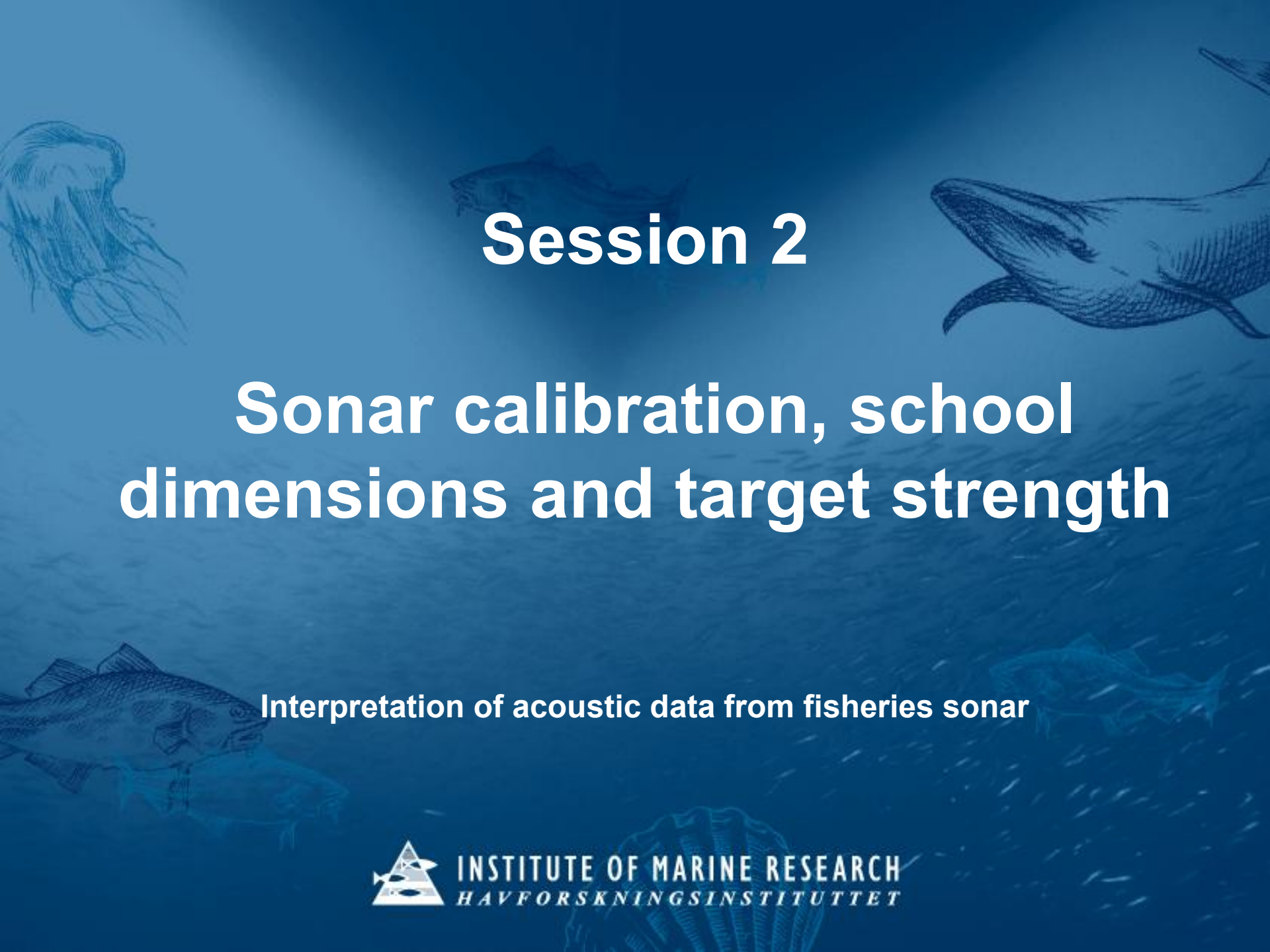




Session 2

Sonar calibration, school dimensions and target strength

Interpretation of acoustic data from fisheries sonar



INSTITUTE OF MARINE RESEARCH
HAVFORSKNINGSINSTITUTTET

Individual school biomass

$$\text{school biomass} = \text{number of fish} * \text{individual fish weight}$$

Sonar data **Trawl data**

$$\text{number of fish} = \text{fish density (n/m}^3\text{)} * \text{school volume (m}^3\text{)}$$

$$\text{fish density} = \frac{\text{volume backscattering coefficient (s}_v\text{, m}^{-1}\text{)}}{\text{backscattering cross section (}\sigma_{bs}\text{, m}^2\text{)}}$$



Individual school biomass

$$\text{number of fish} = \text{fish density } (n/m^3) * \text{school volume } (m^3)$$

$$\text{fish density} = \frac{\text{volume backscattering coefficient } (s_v, m^{-1})}{\text{backscattering cross section } (\sigma_{bs}, m^2)}$$

$$\text{Fish_density} = \frac{s_v}{10^{(TS/10)}}$$

$$s_v = 10^{(S_v/10)}$$

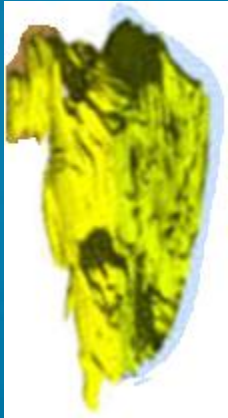
from sonar output we get:
 S_v , mean volume backscattering
strength (dB re 1 m⁻¹)

Side aspect
from in situ or models

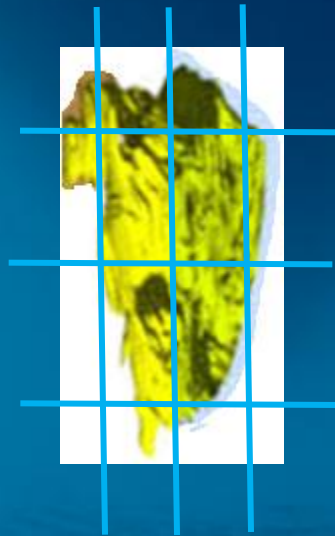


$$\text{number of fish} = \text{fish density (n/m}^3\text{)} * \text{school volume (m}^3\text{)}$$

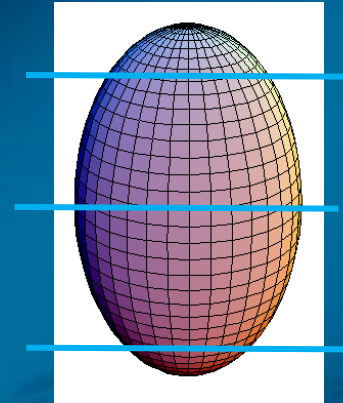
School volume



Real shape from 3D
data of school
morphology
(MS70 multibeam sonar)



Approximation of real shape;
several sonar horizontal and
vertical sections of the school



Approximation to ideal shape
(i.e. spheroid) using only
horizontal sections

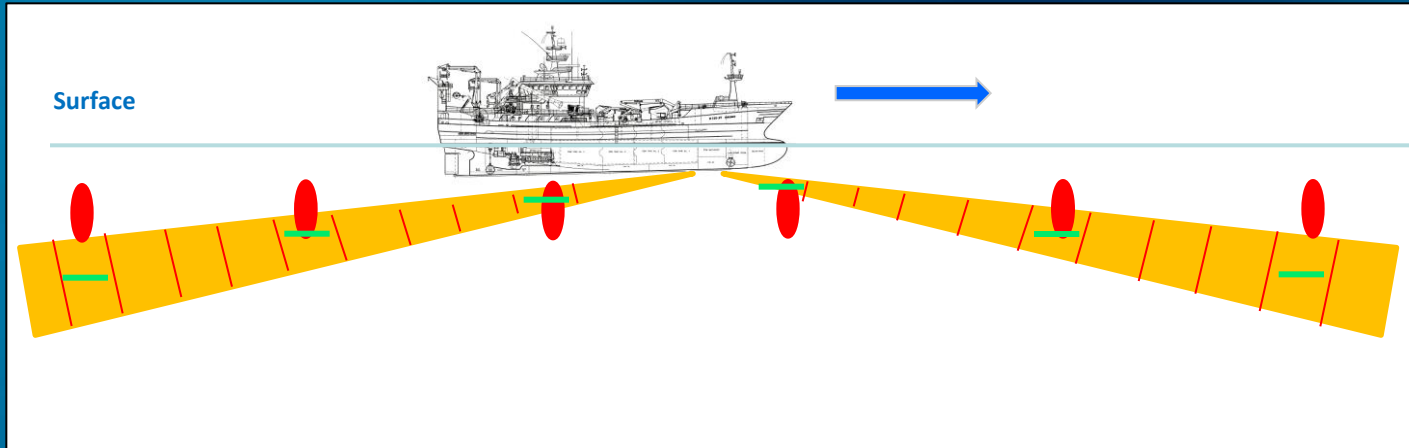


Vertical extension from
echo sounder



$$\text{number of fish} = \text{fish density (n/m}^3\text{)} * \text{school volume (m}^3\text{)}$$

School volume



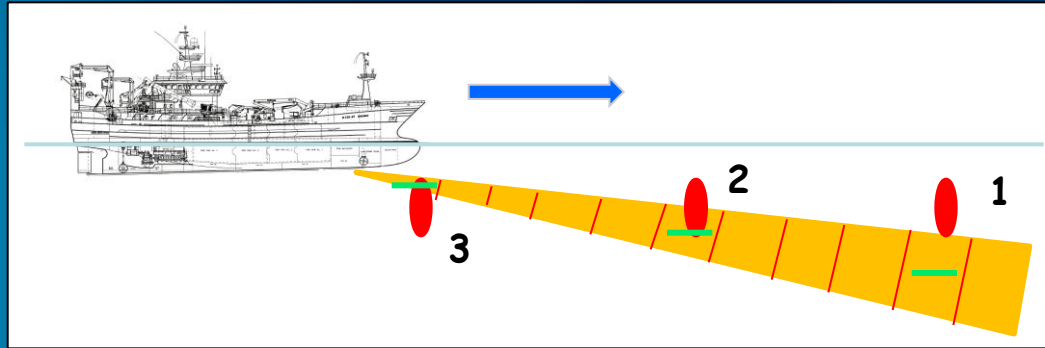
} School real vertical extension
School vertical extension from sonar

School depth derived from sonar provides information about the vertical distribution of schools, **but not accurate** information about **real school depth** or **school vertical extension**

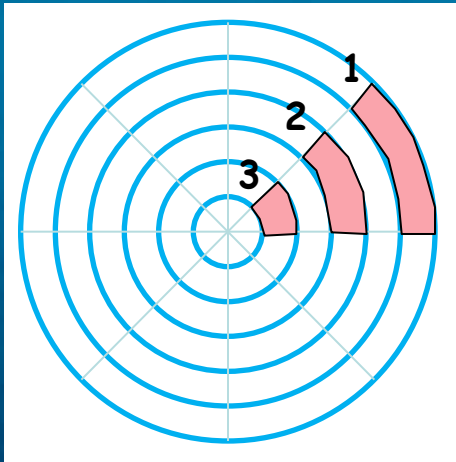


$$\text{number of fish} = \text{fish density (n/m}^3\text{)} * \text{school volume (m}^3\text{)}$$

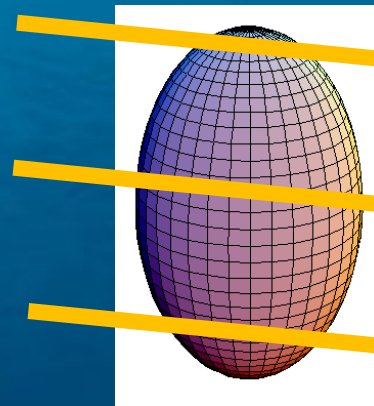
School volume



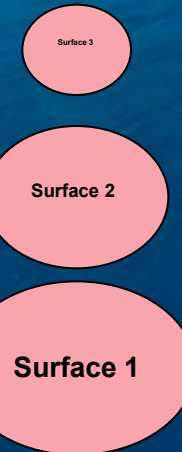
School in horizontal view using omni mode



Sonar sampling at decreasing ranges



School sections from the sonar



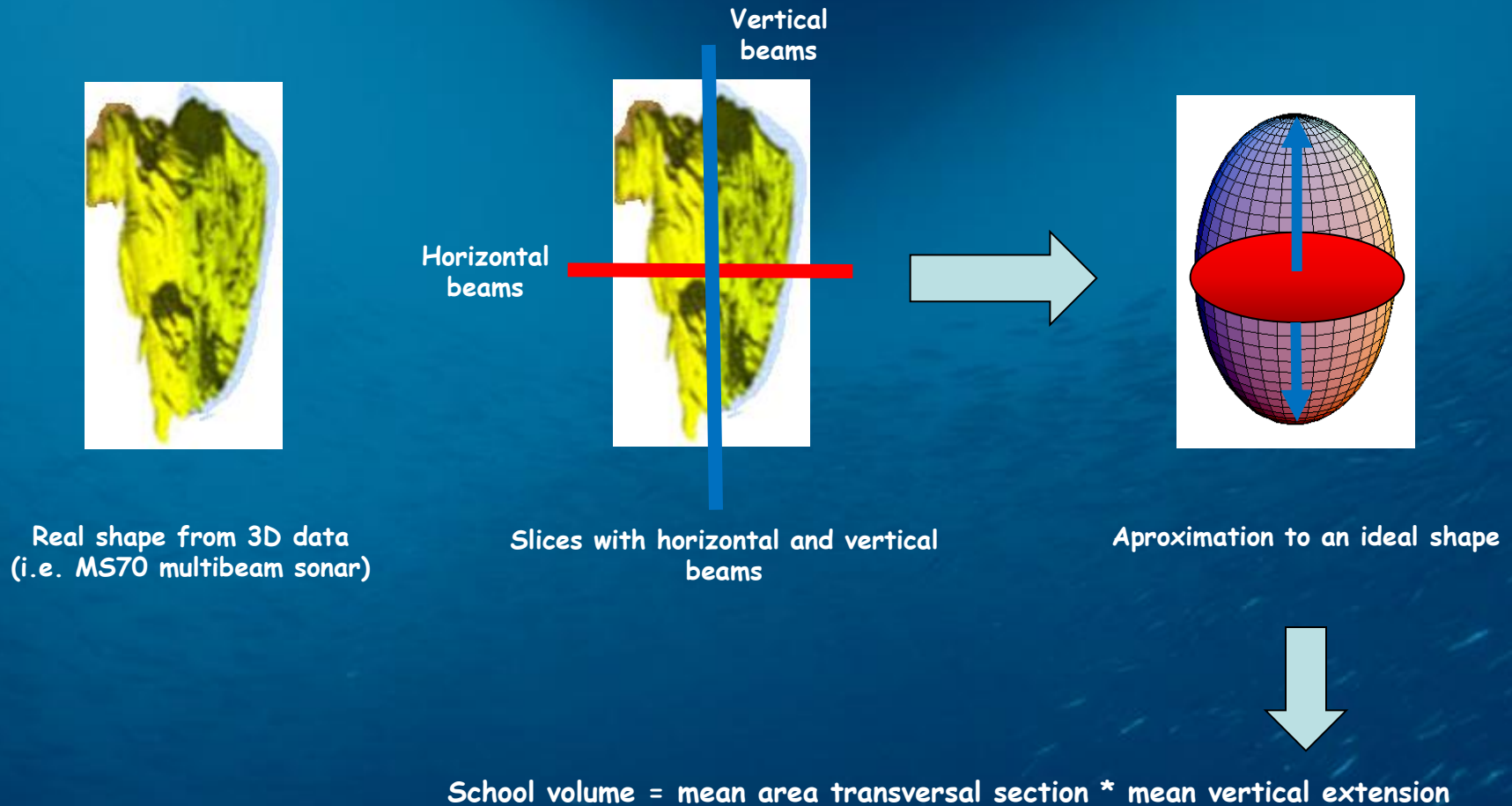
School surface derived from sonar provides rough information about the real school surface which depends on the range. **More accurate at medium distances (2) when school are insonified most**



$$\text{number of fish} = \text{fish density (n/m}^3\text{)} * \text{school volume (m}^3\text{)}$$

School volume

With data from horizontal and vertical beams

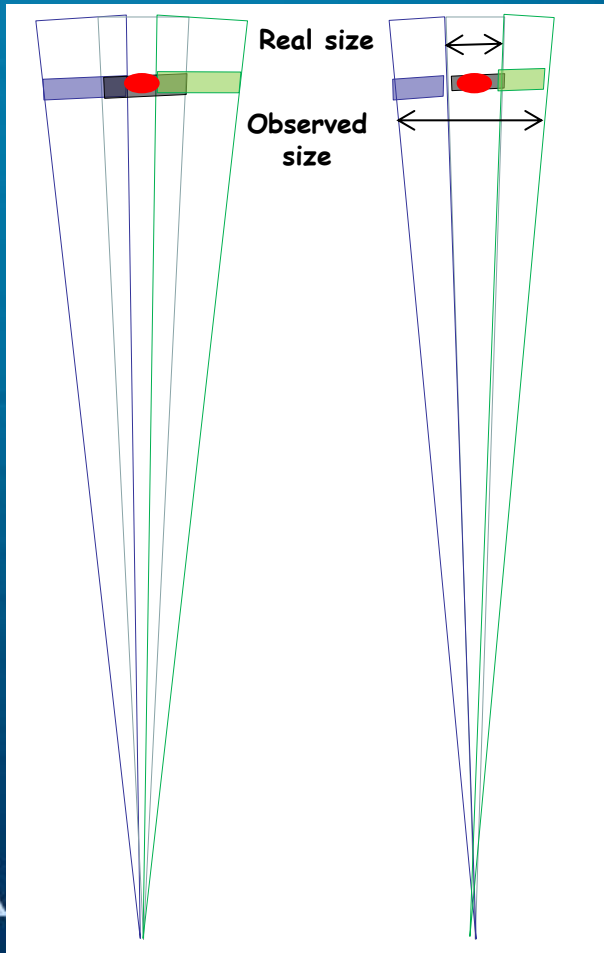


Accurate measurements of school dimensions

Small target

Real overlapping beams
10 deg angle

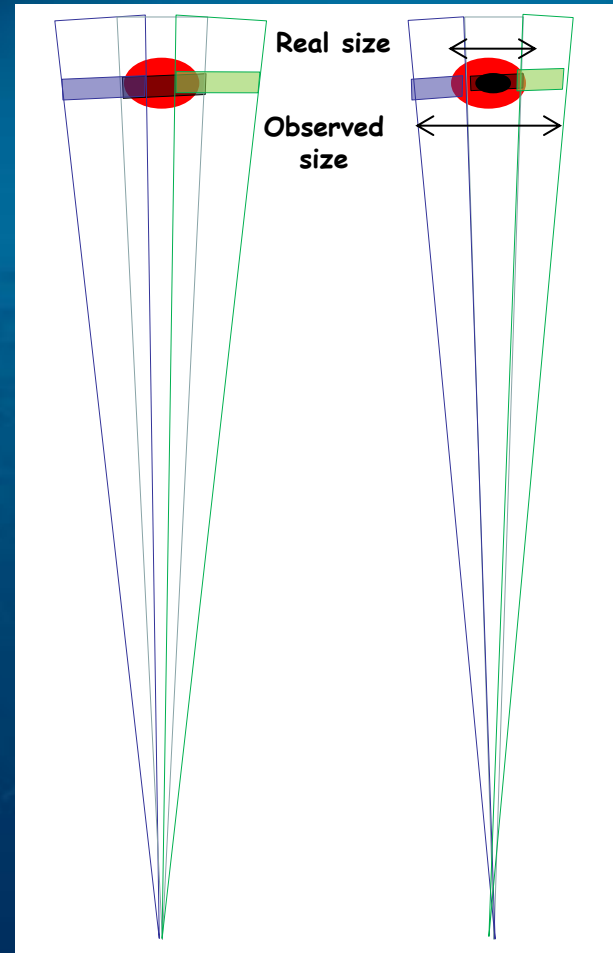
Graphical view
6 deg angle



Large target

Real overlapping beams
10 deg angle

Graphical view
6 deg angle



$$\text{number of fish} = \text{school volume (m}^3\text{)} * \text{fish density (n/m}^3\text{)}$$



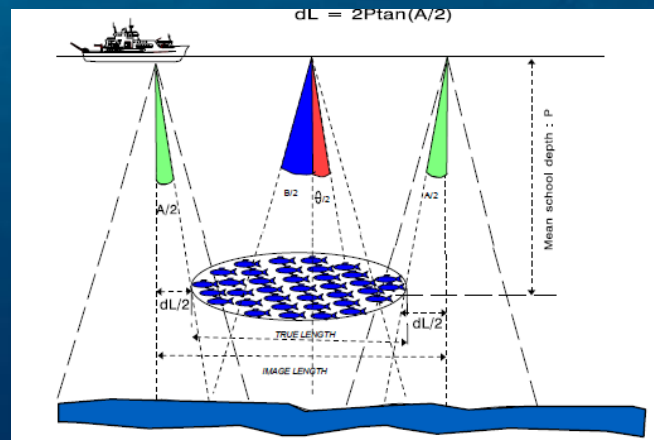
Need accurate measurements
of school dimensions

Correction algorithms (implemented in R):

- Johannesson and Losse, 1977
- Misund, 1990
- Diner, 2001

Limitations:

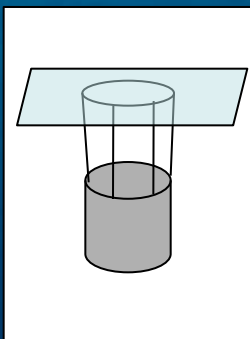
short ranges (<150m)
narrow beams (7-12 deg)
Small/no beam overlap



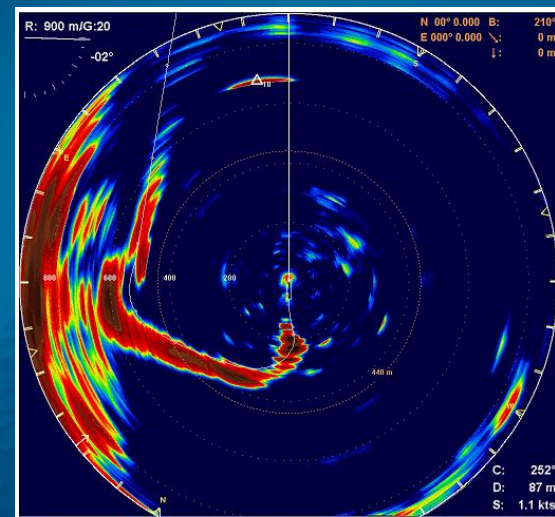
Sonar measurements of known target



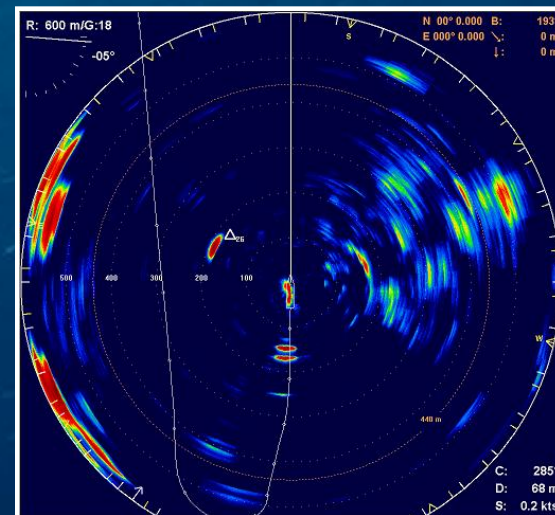
Net pen (13 m diameter) at 15m depth



Net pen at
600 m

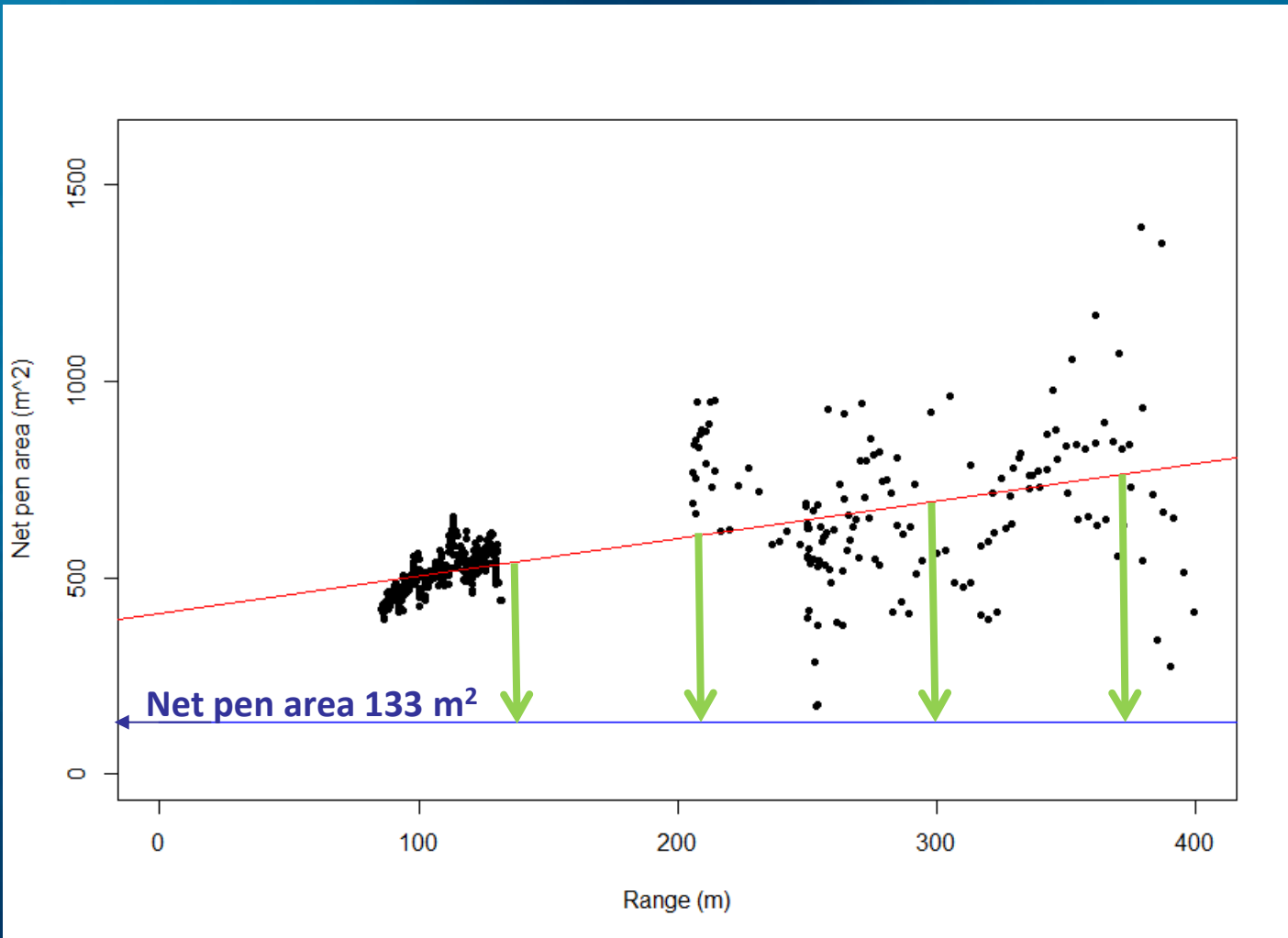


Net pen at
200 m



Accurate measurements of school dimensions

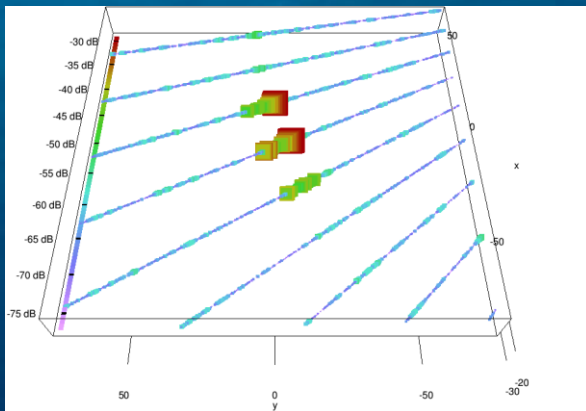
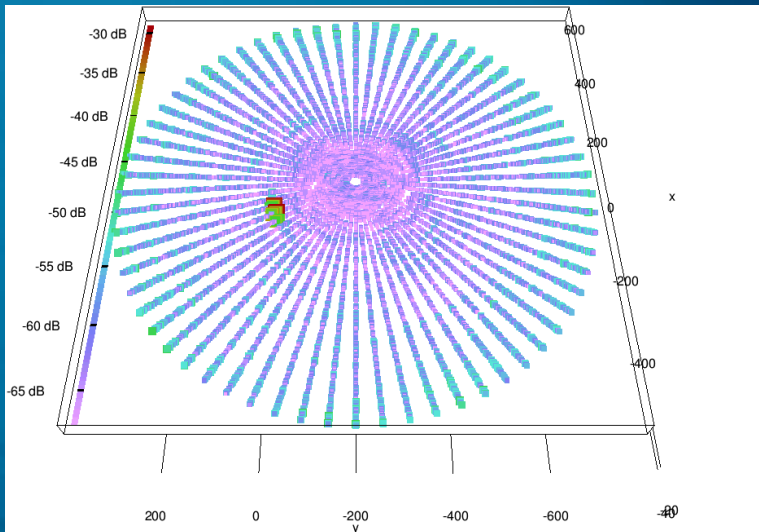
Increase of net pen area (from sonar measurements) with increasing distance from the vessel



Accurate measurements of school dimensions

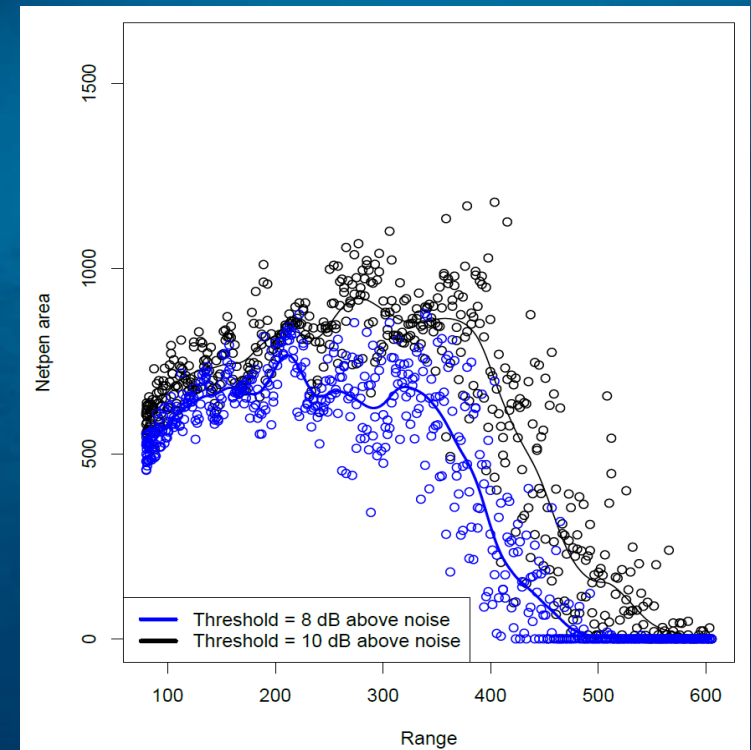
Results of simulations (Arne Holmin)

Simulation of SX90 64 beams and net pen



Detail of net pen (13 m diameter)

Simulated net pen area at variable ranges



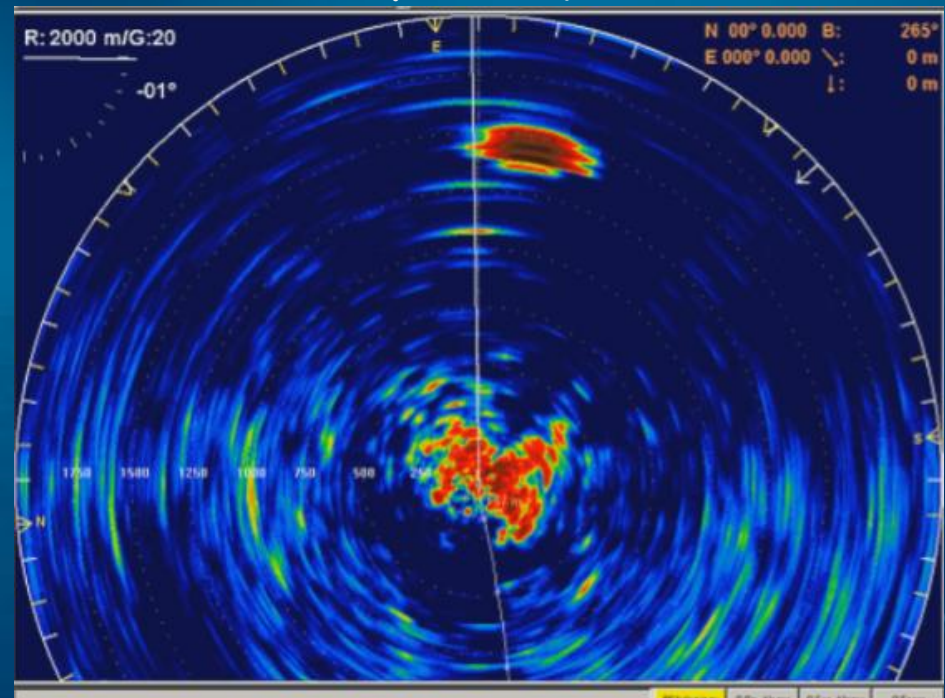
Accurate measurements of school dimensions

Large reference target

Troll A oil platform



Pilars of Troll A in sonar at distance of 2000 m



Improved simulations and new area correction algorithm implemented in Profos



Individual school biomass

$$\text{school biomass} = \underset{\text{Sonar data}}{\text{number of fish}} * \underset{\text{Trawl data}}{\text{individual fish weight}}$$

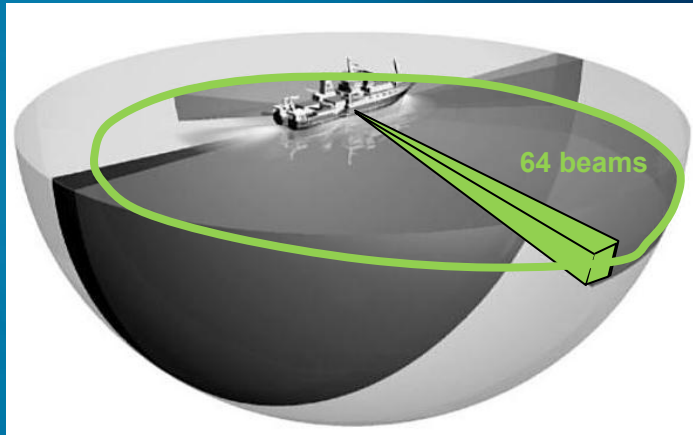
$$\text{number of fish} = \text{fish density (n/m}^3\text{)} * \text{school volume (m}^3\text{)}$$

$$\text{fish density} = \frac{\text{volume backscattering coefficient (s}_v, \text{m}^{-1}\text{)}}{\text{backscattering cross section (}\sigma_{bs}, \text{m}^2\text{)}}$$

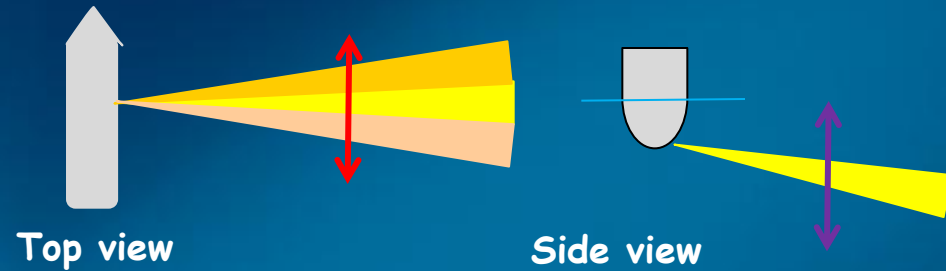
Need accurate s_v



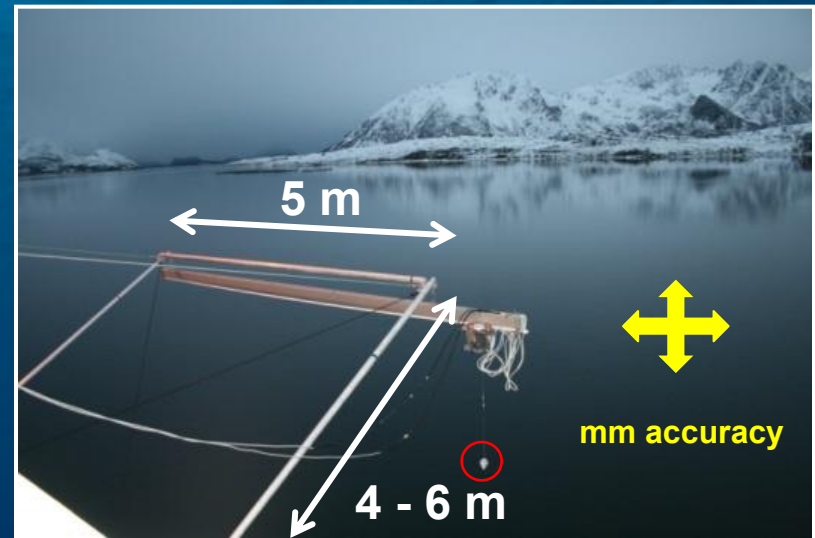
Sonar calibration



Need to move sphere in both horizontal and vertical axis of the beam



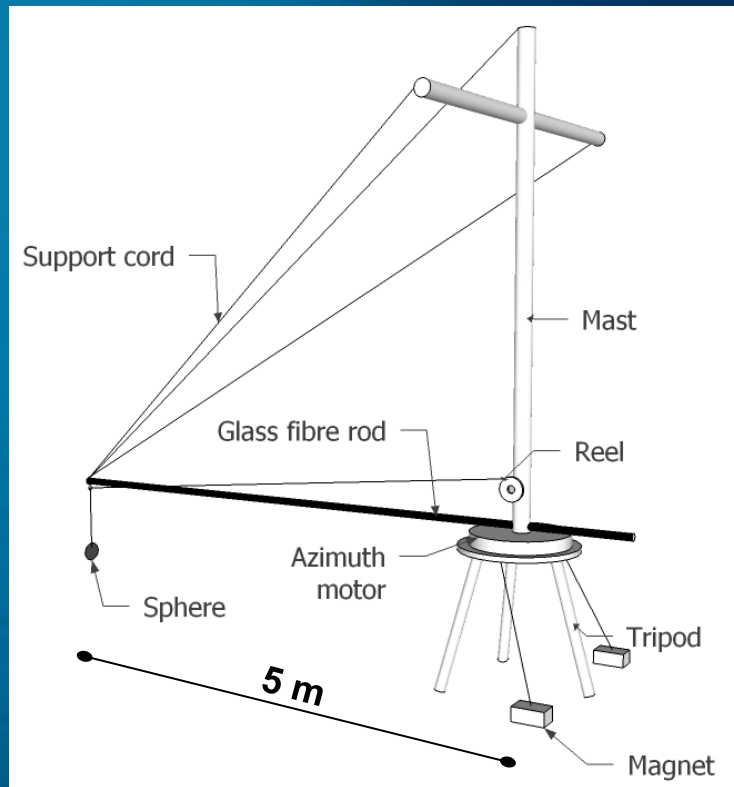
First sonar calibration rig



Remote controlled motors for accurate horizontal and vertical movement



Multi-beam calibration rig



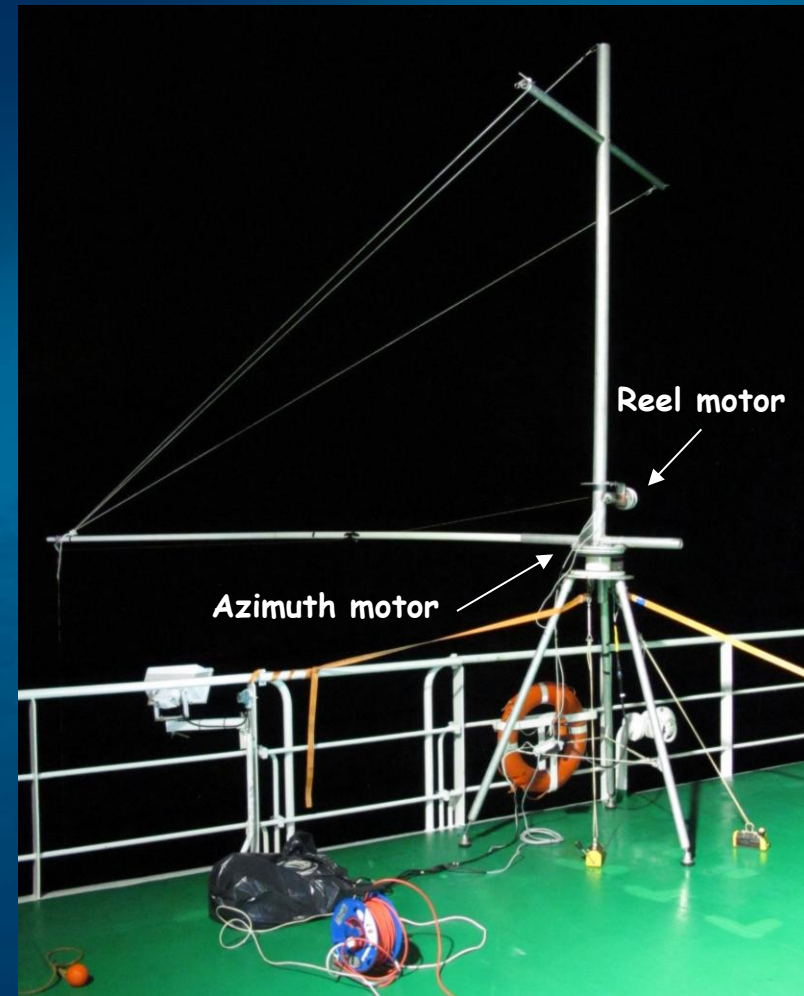
Components:

- Azimuth and reel motors
- Electronic control unit
- System software (via ethernet to the bridge)
- New 64 mm tungsten-carbide calibration sphere



3 persons to mount and move rig
1 hour to mount, 10 min to move

Rig mounted on GO Sars



Two mechanical rigs for field calibration of multi-beam fishery sonars

Sindre Vatnehol*, Atle Totland, Egil Ona

Methods in Oceanography 13–14 (2015) 1–12

New rig made at Valencia Polytechnic University (2021)

«Off the shelf» product



Sonar calibration in Tarragona, Spain



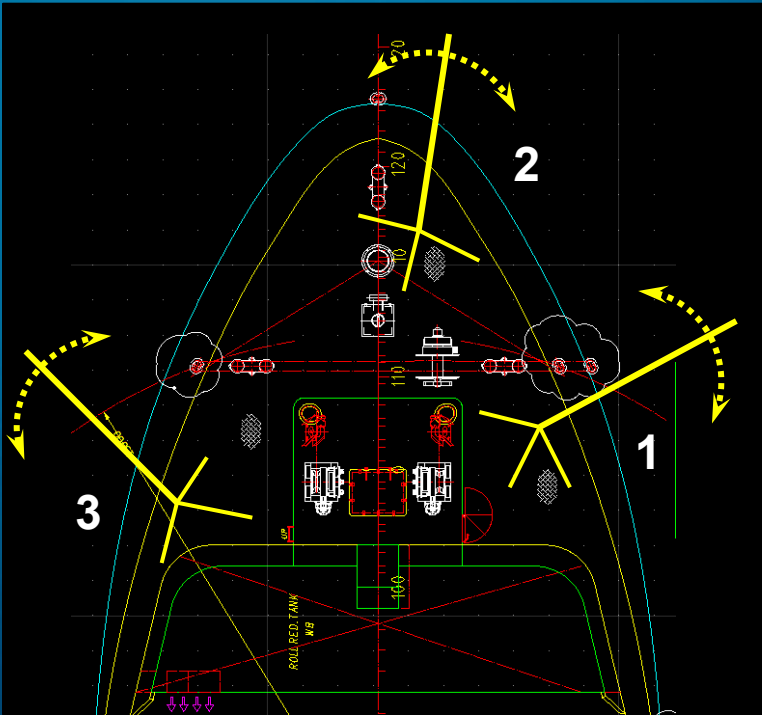
Components:

- Aluminium
- Azimuth motors in base (lighter + stable)
- WiFi connection to bridge
- Mobile app for operation (feedback azimuth and depth)



Calibration

Arrangement of rig in three positions onboard FV Kings Bay



Methods in Oceanography

Volume 17, December 2016, Pages 206–220



Practical calibration of ship-mounted omni-directional fisheries sonars

Gavin J. Macaulay^a, Sindre Vatnehol^a, Ole Bernt Gammelsæter^b, Héctor Peña^a, Egil Ona^a



Vessels used and sonars calibrated

R/V "GO Sars", SX+SU90



F/V "Artus", SX90



F/V "Kings Bay", SU90



F/V "Eros", SU90



F/V "Brennholm", SX90



R/V "Fridtjof Nansen", SU90



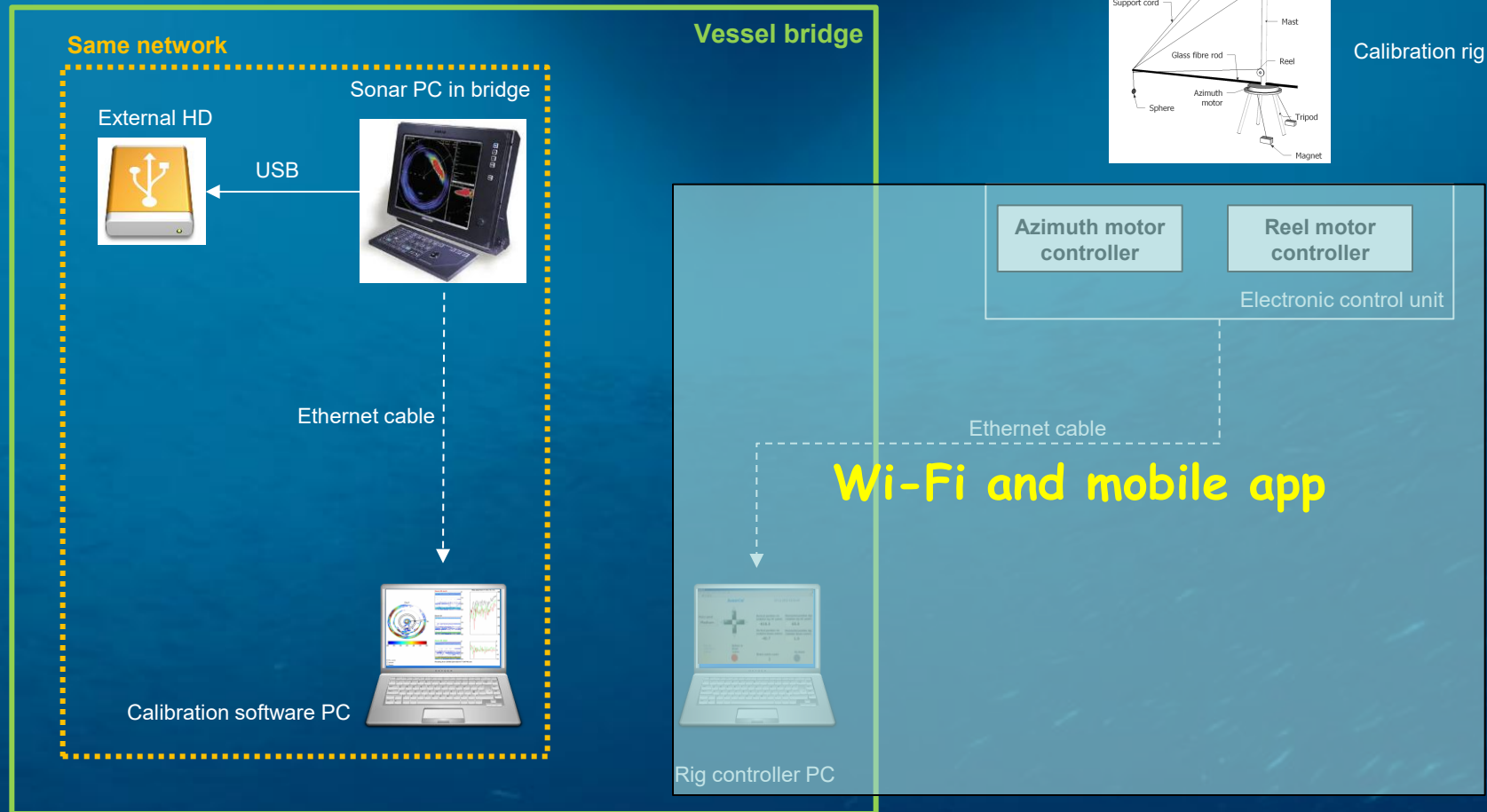
Summary of calibrations

Ship	Calibration date	Sonar	Beam mode	Frequency (kHz)	Pulse type
GO Sars	10.11.2012	SX90	horizontal	26	FM
Artus	16.03.2013	SX90	horizontal	30	CW
GO Sars	17.03.2013	SX90	horizontal	26	CW
Artus	07.11.2013	SX90	horizontal	30	CW
GO Sars	10.11.2013	SX90	horizontal	26	CW
Kings Bay	20.10.2014	SU90	horizontal	30	CW
GO Sars	21.10.2014	SX90	horizontal	26	CW
Eros	19.10.2015	SU90	horizontal	30	FM
GO Sars	20.10.2015	SU90	horizontal	26	FM
Kings Bay	02.05.2016	SU90	horizontal	30	FM
Brennholm	19.09.2016	SU90	horizontal	30	FM
GO Sars	28.10.2016	SU90	horizontal	26	FM
Nansen	01.02.2017	SU90	horizontal	26	FM
Nansen	01.02.2017	SH90	horizontal	114	FM
Eros	13.02.2017	SU90	horizontal	30	FM
Vendla	13.02.2017	SX90	horizontal	30	FM
Kings Bay	13.02.2017	SU90	horizontal	30	FM
GO Sars	03.05.2017	SU90	horizontal	26	FM
Eros	06.11.2017	SU90	horizontal	30	FM
Eros	06.11.2017	SU90	vertical	30	FM
GO Sars	20.11.2017	SU90	horizontal	26	FM
Vendla	13.02.2018	SX90	horizontal	30	FM
Eros	13.02.2018	SU90	horizontal	30	FM
Kings Bay	13.02.2018	SU90	horizontal	30	FM
Johan Hjort	29.06.2017	SU90	horizontal	26	FM
GO Sars	30.04.2018	SU90	horizontal	26	FM
Johan Hjort	29.06.2018	SU90	horizontal	26	FM
Kronprins Haakon	17.07.2018	SU90	horizontal	26	FM
Kronprins Haakon	18.07.2018	SH90	horizontal	114	FM

Until February 2019: 9 vessels, 28 calibrations, 288 beams

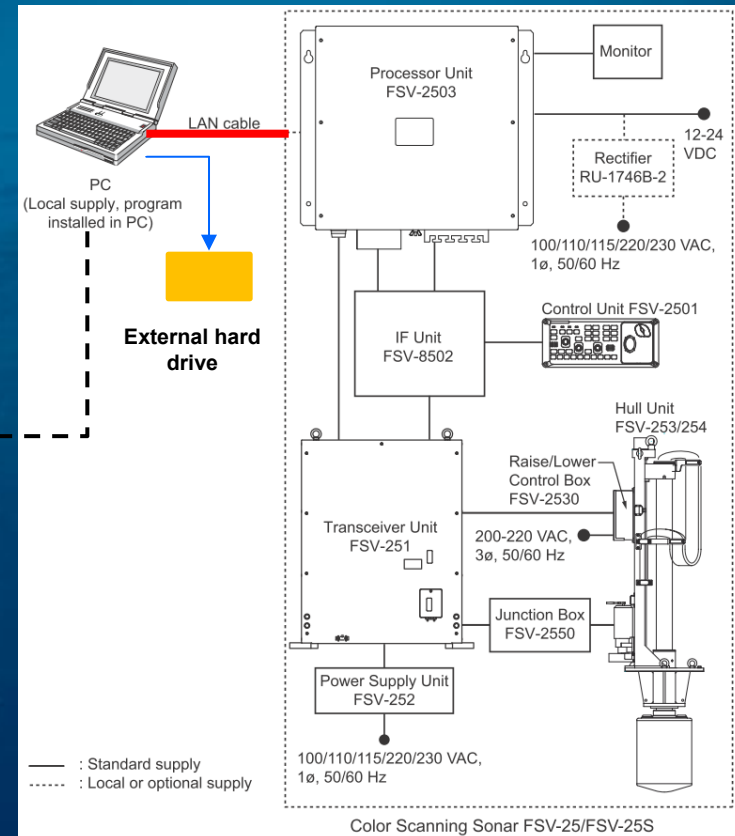


Calibration elements (Simrad)

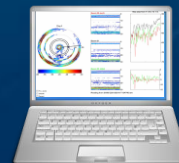


System configuration for data storage

FSV-25 configuration diagram enabled for netCDF data recording

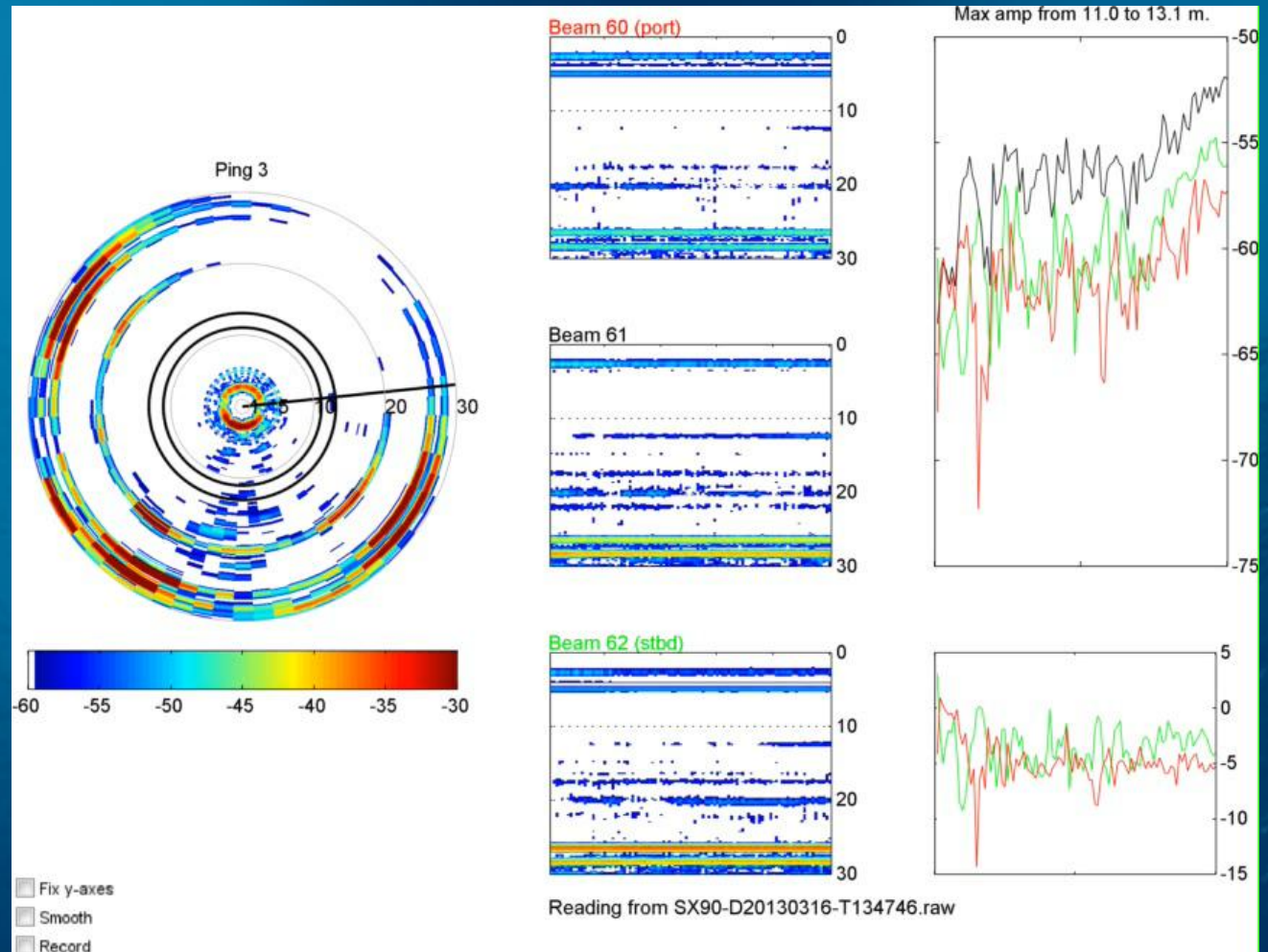


Calibration software PC



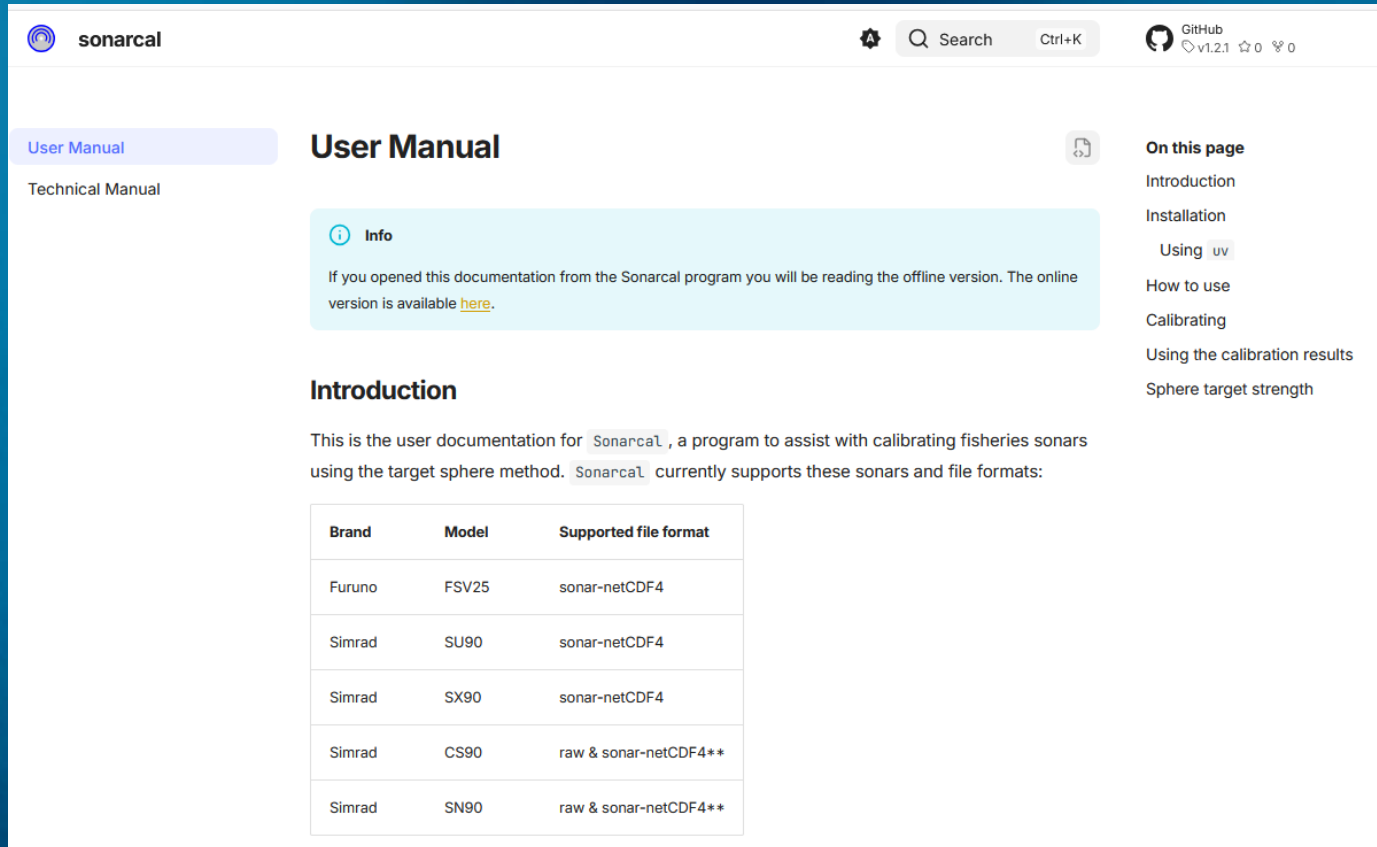
Calibration procedure

Custom written software to read raw data during calibration



Calibration procedure

Python code free available for sonar calibration (Gavin Macaulay_Aqualyd / AZTI)



The screenshot shows the GitHub repository page for 'sonarcal'. The page title is 'User Manual'. On the left, there is a sidebar with 'User Manual' and 'Technical Manual' links. The main content area has a light blue information box stating: 'If you opened this documentation from the Sonarcal program you will be reading the offline version. The online version is available [here](#).' Below this is the 'Introduction' section, which states: 'This is the user documentation for Sonarcal, a program to assist with calibrating fisheries sonars using the target sphere method. Sonarcal currently supports these sonars and file formats:'. A table follows, listing supported sonar models and their file formats.

Brand	Model	Supported file format
Furuno	FSV25	sonar-netCDF4
Simrad	SU90	sonar-netCDF4
Simrad	SX90	sonar-netCDF4
Simrad	CS90	raw & sonar-netCDF4**
Simrad	SN90	raw & sonar-netCDF4**

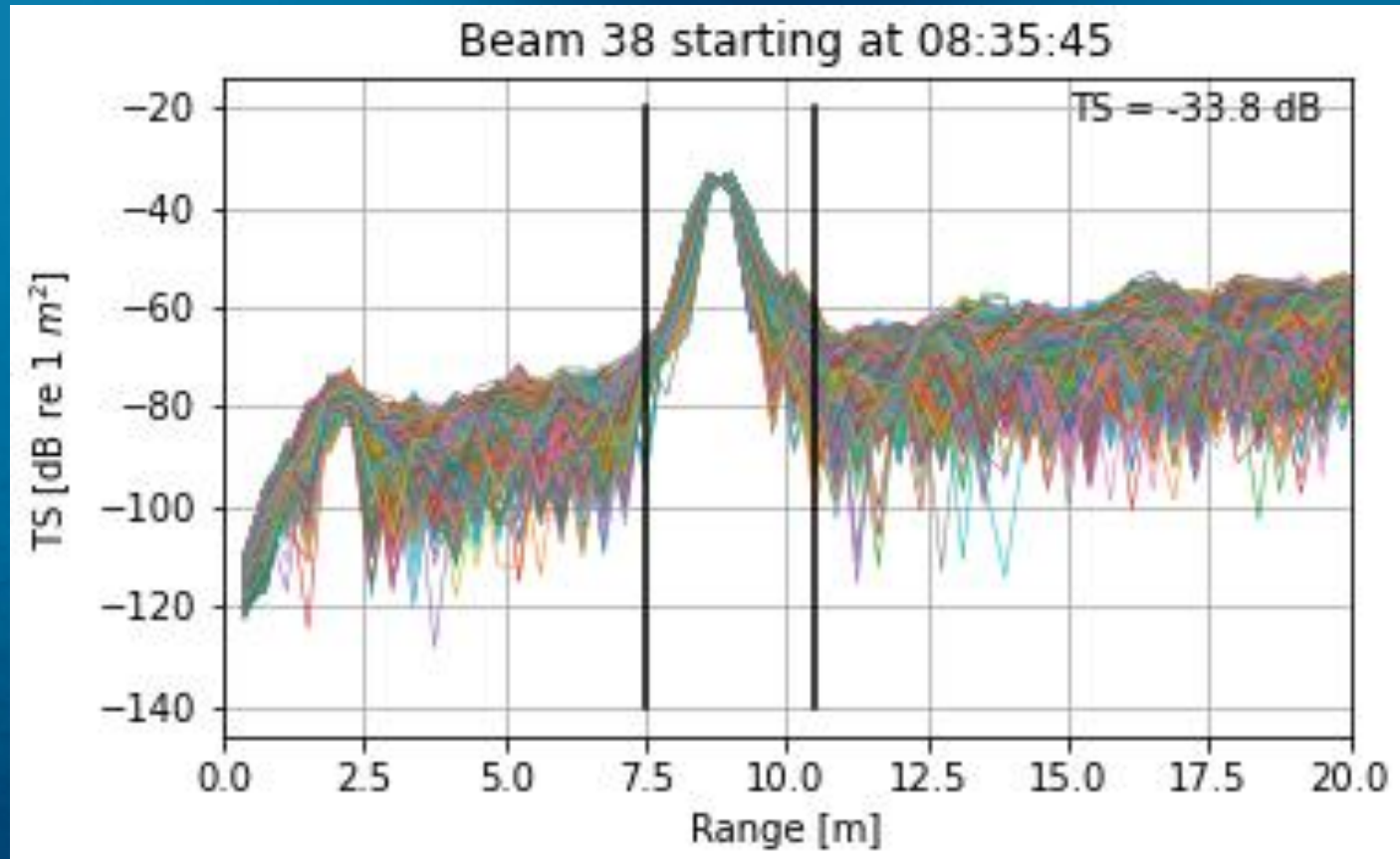
On the right side of the page, there is a 'On this page' section with links to: Introduction, Installation, Using uv (highlighted), How to use, Calibrating, Using the calibration results, and Sphere target strength.

<https://aqualyd-limited.github.io/sonarCal/>



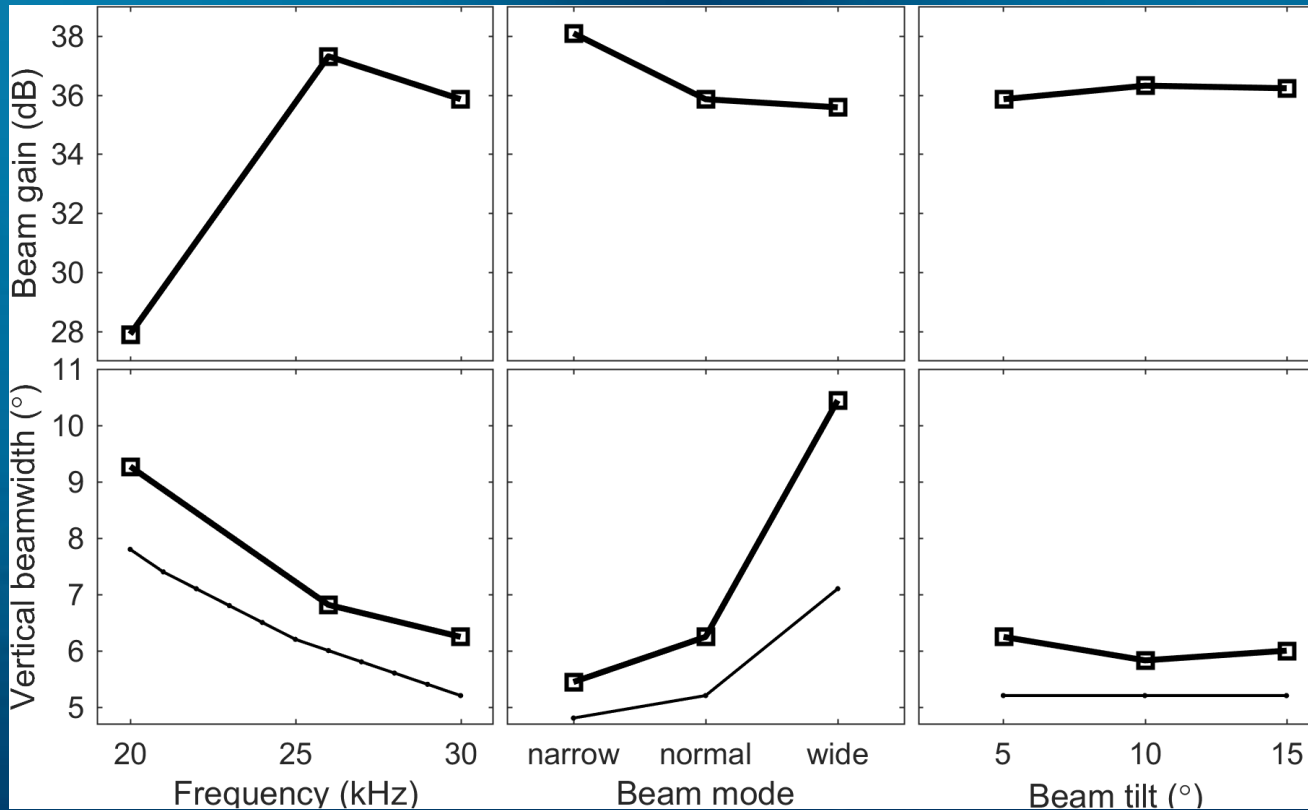
Calibration procedure

Measurements of calibration sphere TS



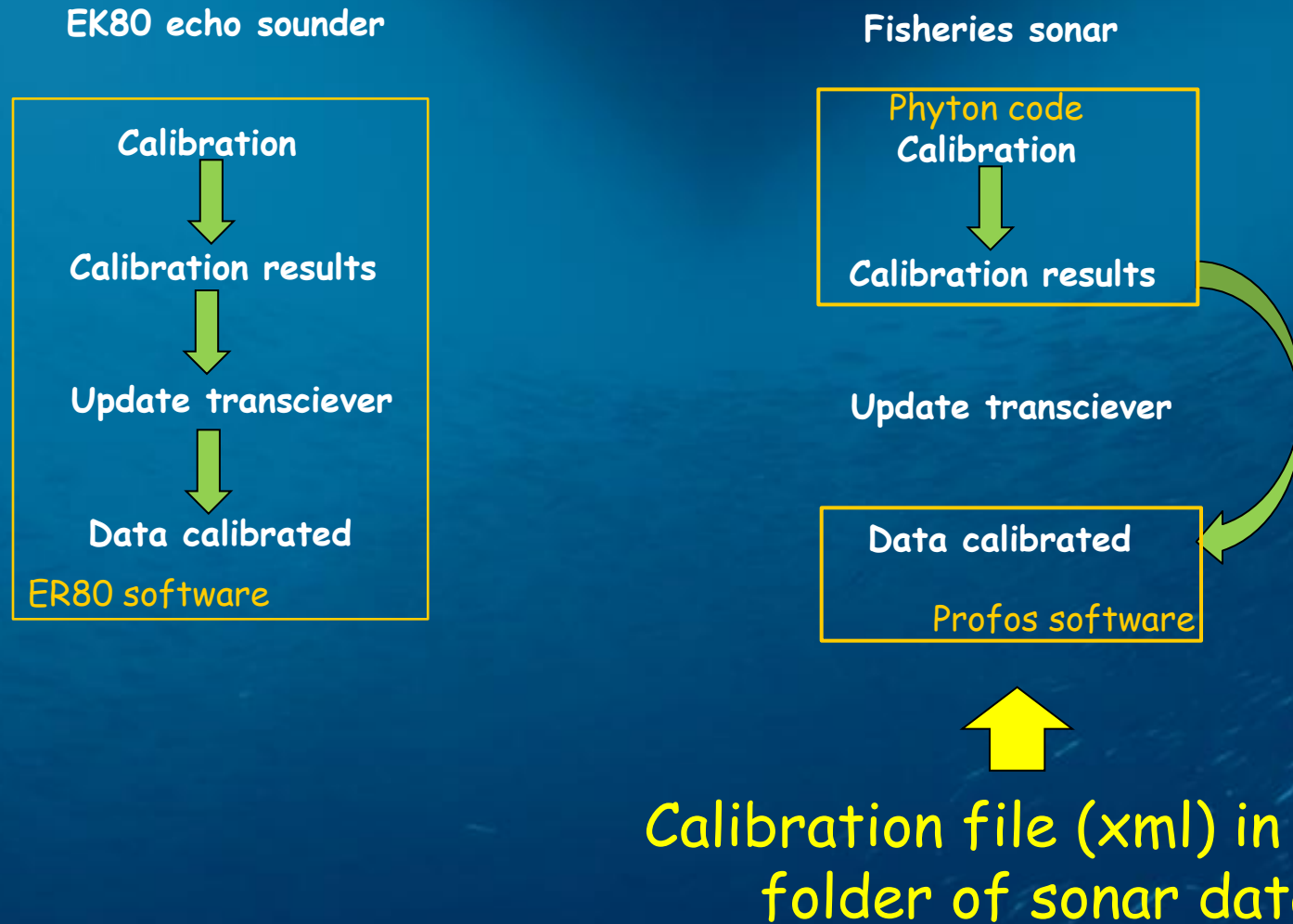
Results

Gain and vertical beam width for different sonar parameters
SU90 Eros (—) and predicted by manufacturer (—)



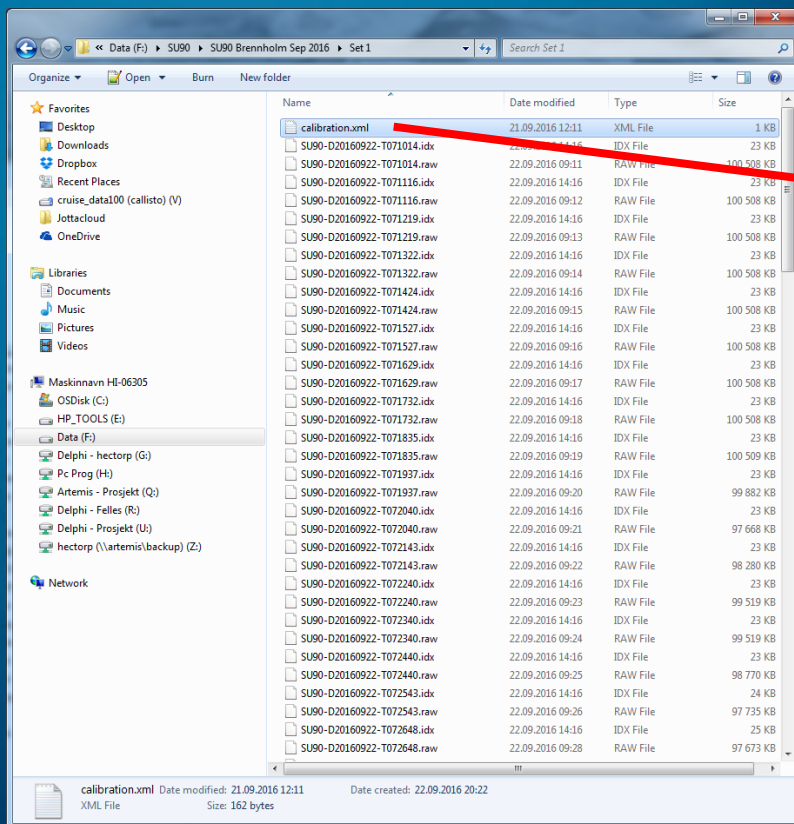
Sonar calibration

Data correction using calibration results



Sonar calibration

Calibration file (xml) in same folder of sonar data



calibration.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<calibration>
  <default g="38" ro="3" tauEff="0.0019" tau="0.004" frequency="30000"
  beamWidthMode="0"/>
</calibration>
```



RAW and netCDF data formats

Simrad sonar RAW data format

- Only sonar producer with scientific output available (2004)
- Proprietary data format for internal company use
- Limited documentation + no update of changes
- Required complex code to read



Interest of Furuno to implement scientific output in low frequency sonars (2016)



Need for a standard documented data format



Proposed to ICES WGFASST to create a "Topic group" to define a sonar data format



netCDF (network Common Data Form) data format



Publisehd May 2018

This report documents a convention for the storage of sonar data in netCDF4-formatted computer files, with an initial focus on omnisonars. The intention is to provide a well-founded convention that is supported by multiple sonar systems and multiple sonar analysis software packages, with the aim of facilitating the use of sonar data for research and survey purposes. The name of this convention is SONAR-netCDF4.

- ability to adequately represent the content and structure of sonar data and associated metadata
- standardized, open-file format
- fast random access to data stored in the file
- ability to store multiple types of data (e. g. position and backscatter)
- ability to store metadata (e.g. sonar settings)
- **freely available and open libraries to read the data files** into programming languages and technical computing environments (e.g. Java, C, C++, Python, Matlab, and R)
- long-term support and extensive use in other scientific fields
- support for very large datasets

**Includes equations for Sv and TS for Simrad (Type 1)
and Furuno (Type 2) sonars**



Manipulate netCDF data files

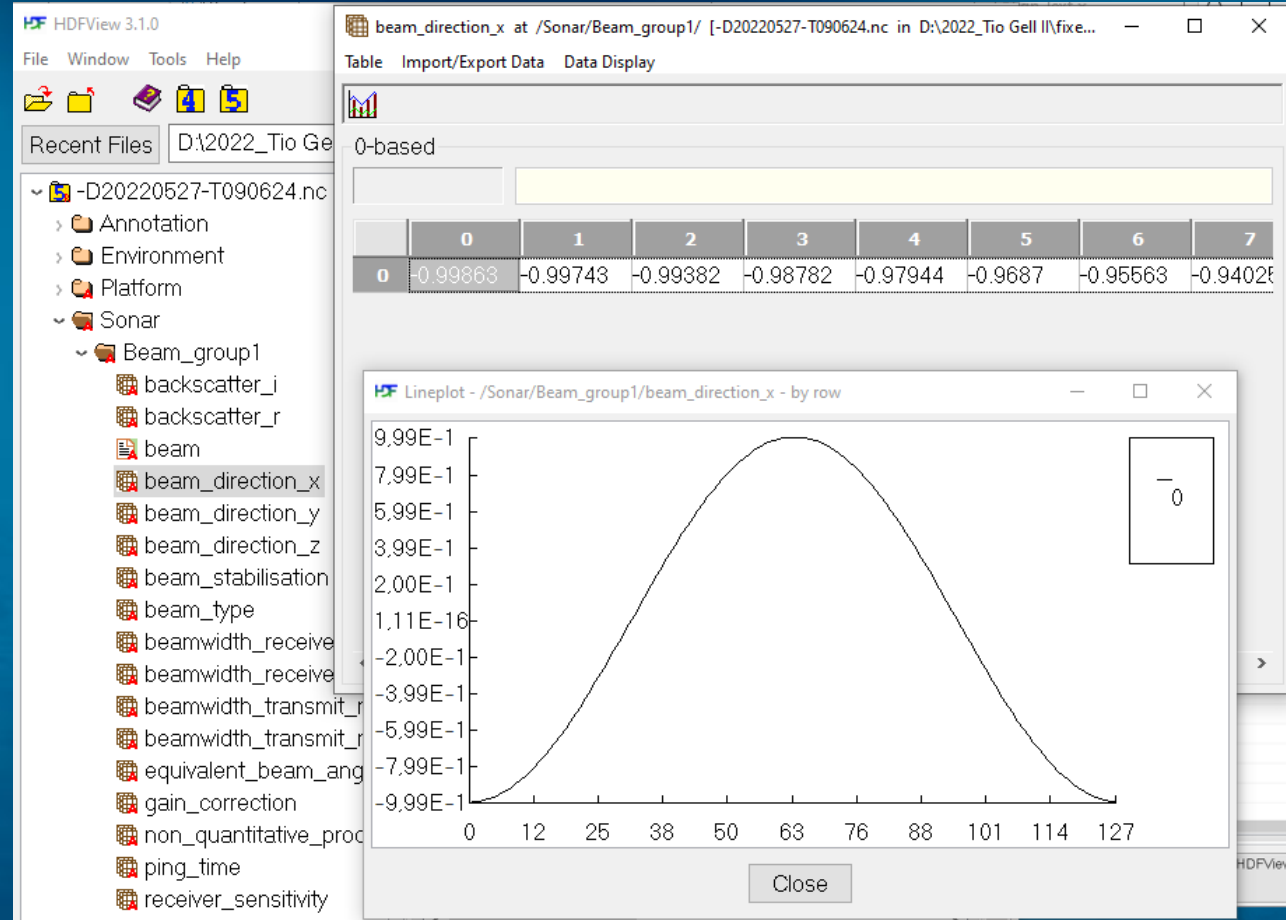
Free software: HDFView

R-package: ncdf4

Python packages: netCDF4, gdal



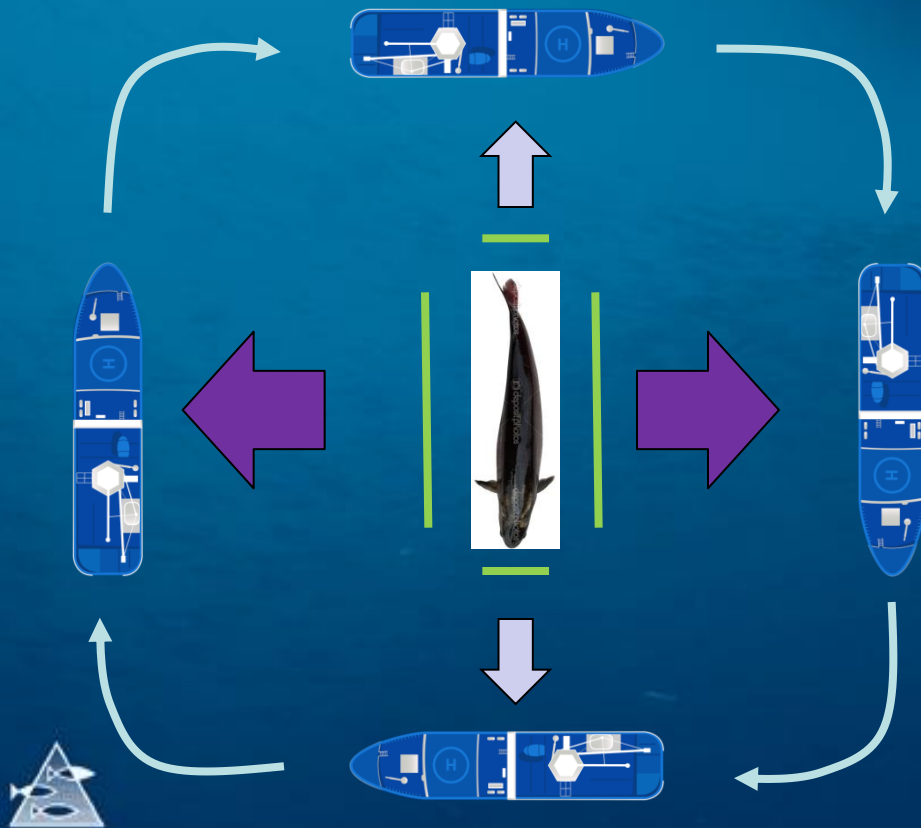
1 line code to add
calibration parameters
to a file



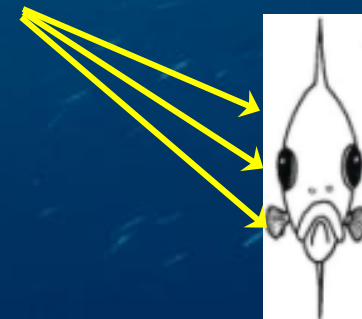
fish density = $\frac{\text{volume backscattering coefficient } (s_v, \text{m}^{-1})}{\text{backscattering cross section } (\sigma_{bs}, \text{m}^2)}$

backscattering cross section (σ_{bs} , m^2)

Need side aspect target strength



- Strong backscattering from side aspect
- Low backscattering from head or tail

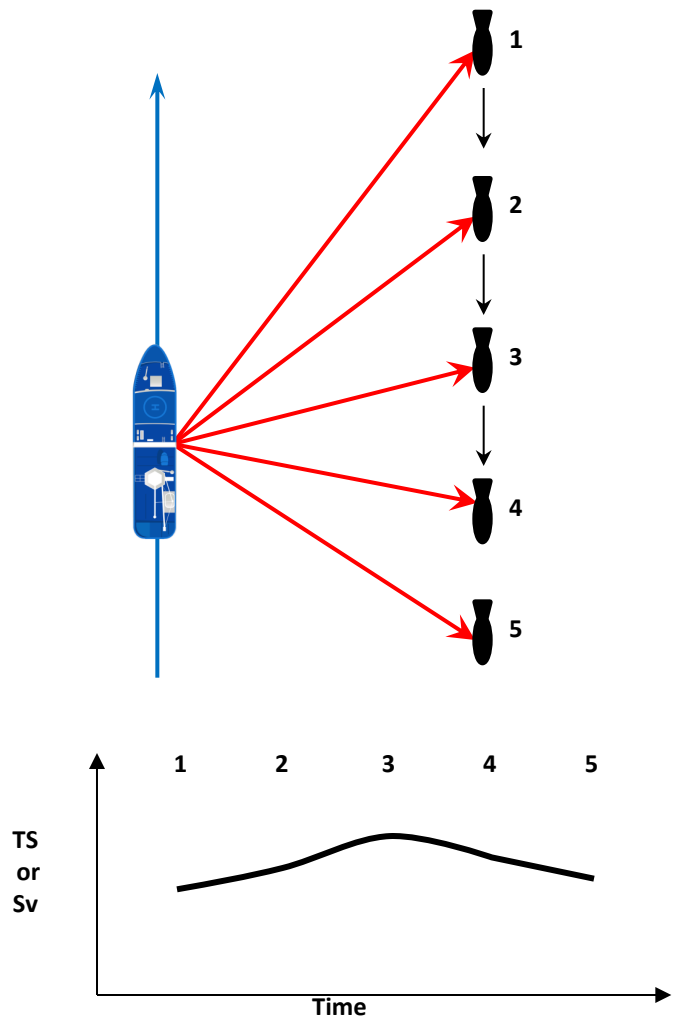


Smaller changes at different tilt angles

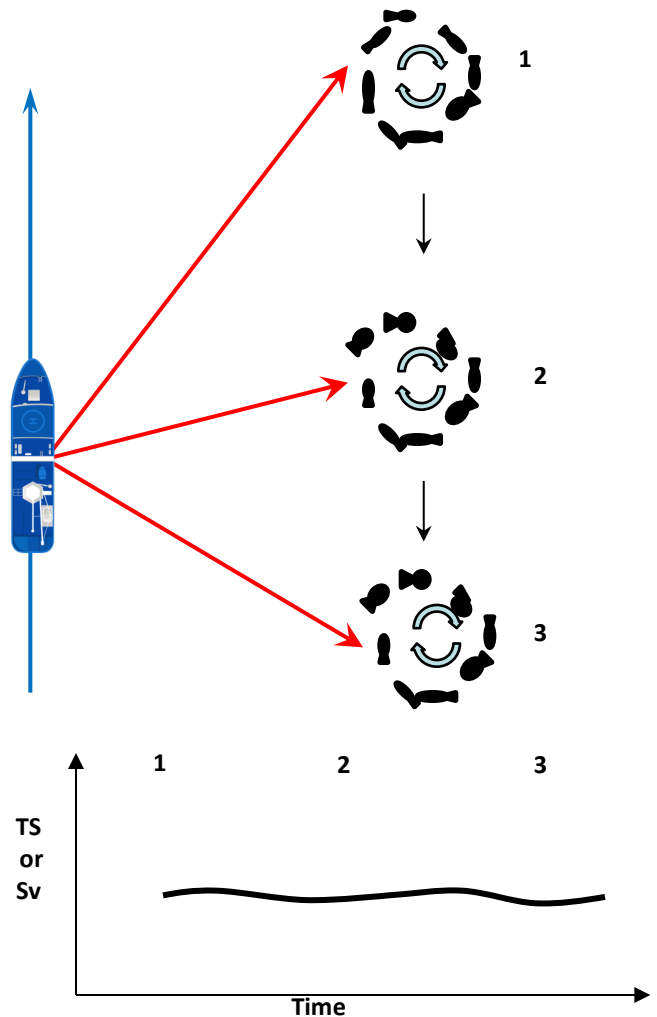
Side aspect target strength

Insonification angle from sonar to school

Surveying on polarized schools



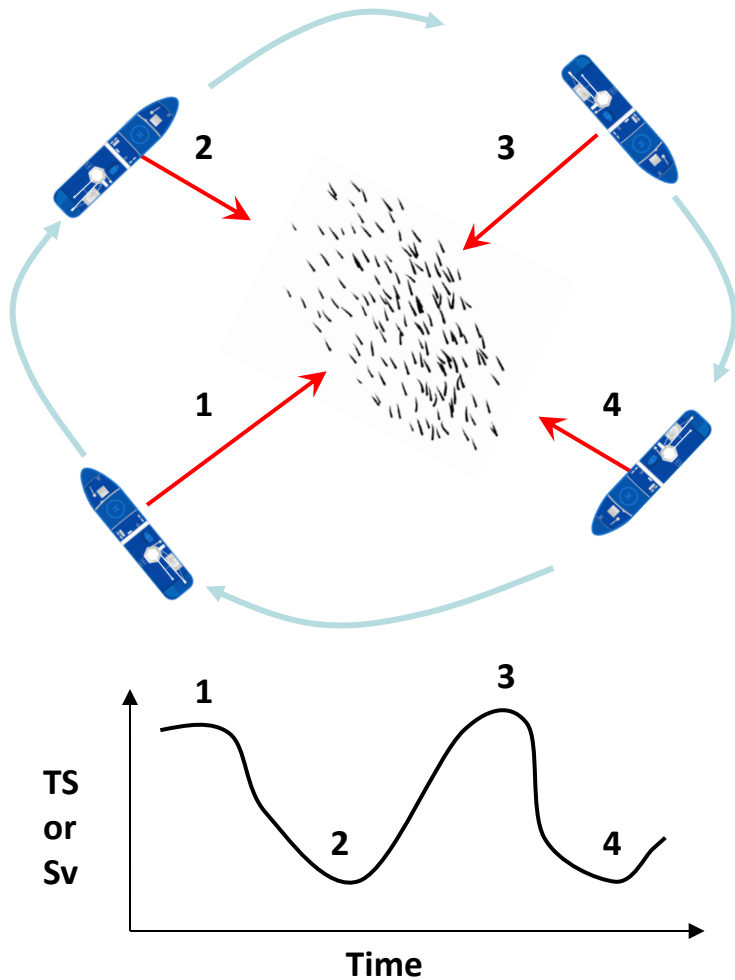
Surveying on milling schools



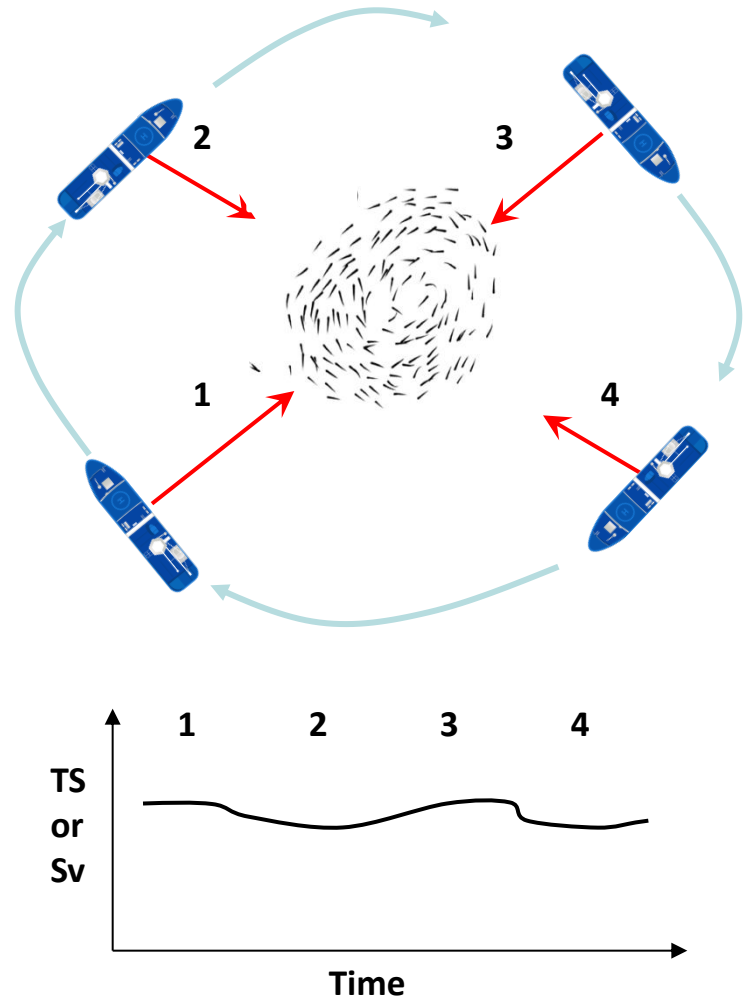
Side aspect target strength

Insonification angle from sonar to school

Detailed inspection on polarized schools



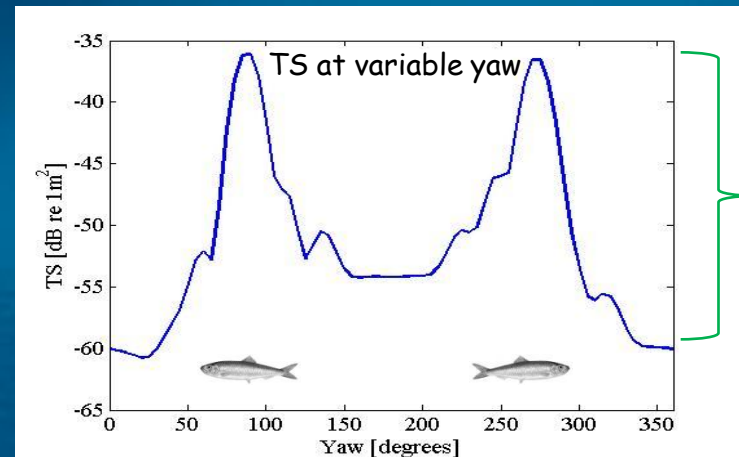
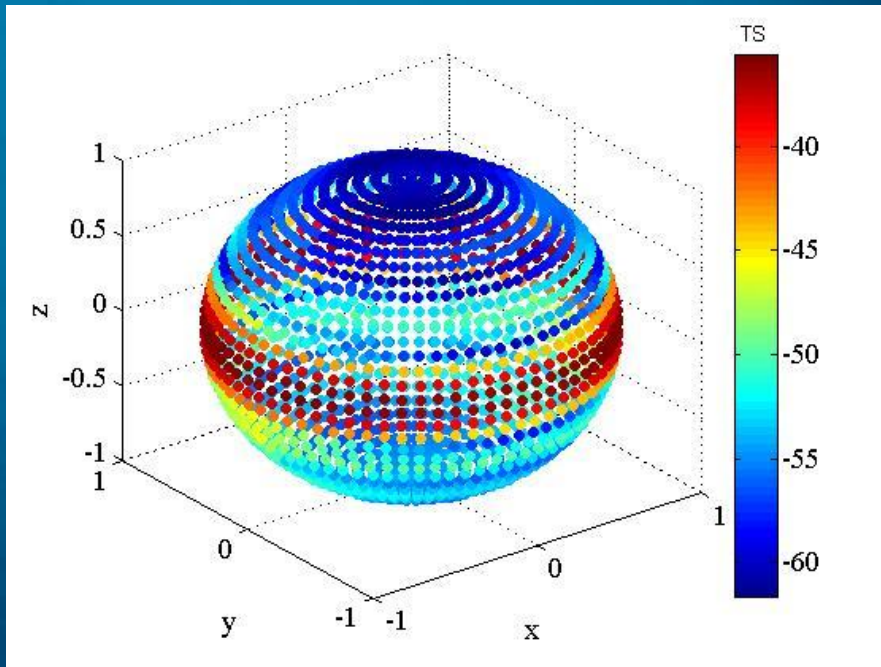
Detailed inspection on milling schools



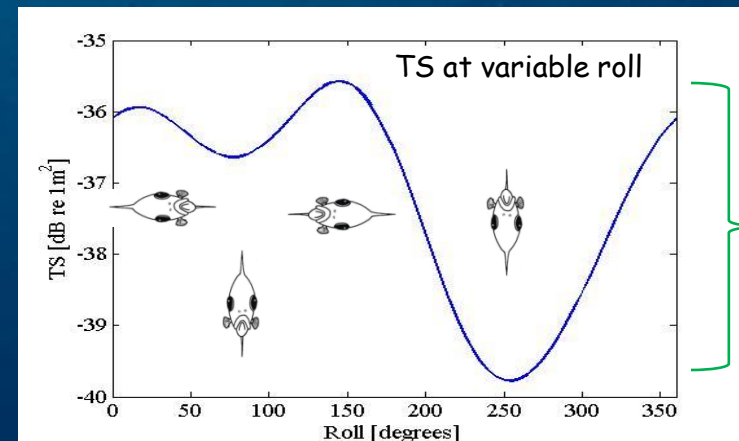
Side aspect target strength

Modelled TS of herring swimbladder at variable yaw

Target strength in any aspects (5 deg resolution)



20 dB difference

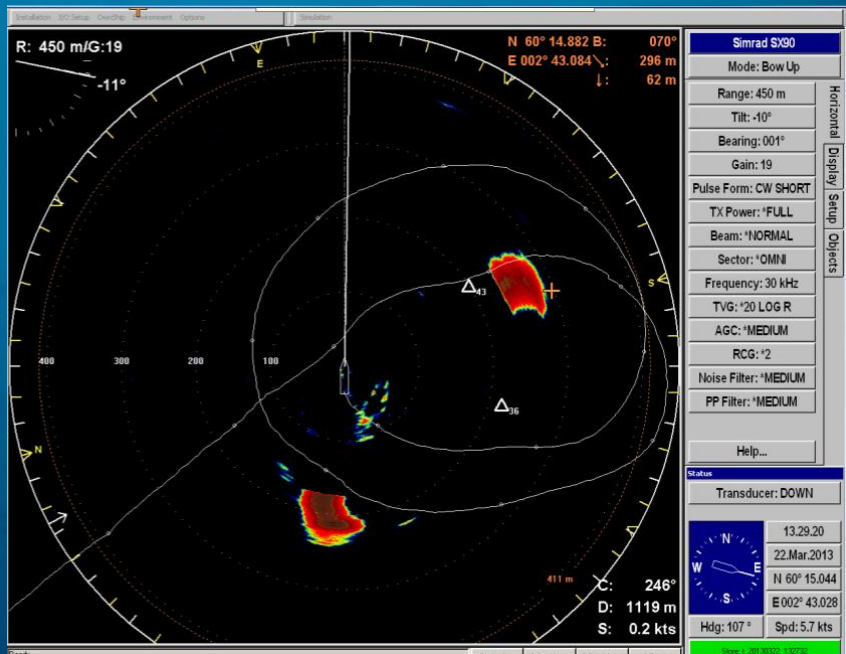


4 dB difference

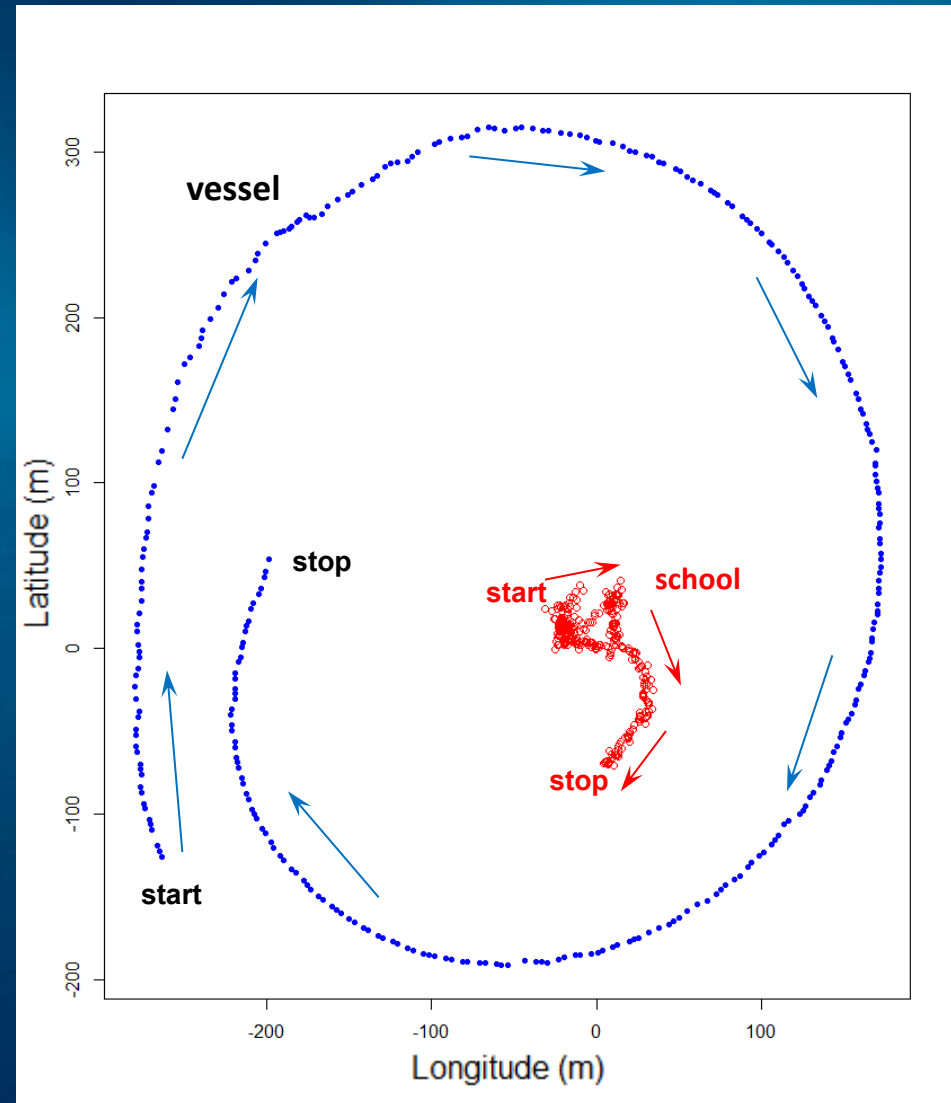


Side aspect target strength

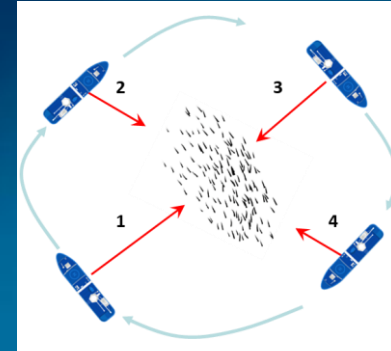
Detailed inspection of single herring school



Sonar image



Side aspect target strength



Herring school Sv and vessel dynamics during detailed inspection

